1 Homework

1.1 Peak Value and Effective Value

I_0 and U_0 are the peak values, representing the highest current and voltage reached during one cycle.
The effective values are given by:
$U_{\text{eff}} = \frac{U_0}{\sqrt{2}}$ and $I_{\text{eff}} = \frac{I_0}{\sqrt{2}}$.
These effective values allow a direct comparison between alternating current and direct current.

1.2

Fig. 6.a

$$\frac{U_{\text{out}}}{U_{\text{in}}} = \frac{1}{\omega C i\left(R + \frac{1}{\omega C i}\right)} = \frac{1}{i\omega R + 1} < 1 \Rightarrow \text{low-pass-filter} \tag{1.1}$$

Fig. 6.b

$$\frac{U_{\text{out}}}{U_{\text{in}}} = \frac{i\omega L}{R + i\omega L} = \frac{1}{\frac{R}{i\omega L} + 1} = \frac{1}{1 - \frac{Ri}{\omega L}} > 1 \Rightarrow \text{high-pass-filter} \tag{1.2}$$

2 Part I - Frequency-dependent voltage divider and filter

In the first experiment we made a Low pass filter with a capacitor as illustrated in Figure 1. The combination of capacitor (C) and resistor (R) used in the circuit leads to different effects, which we want to observe in this experiment. More specifically, we look at the ration of $U_{\text{out}}/U_{\text{in}}$ and make a graphical representation. Then compare the cut-off-frequency and some limit cases. Will talk about what the roll-off is and finally what kind of face shift between U_{out} and U_{in} this circuit results in.

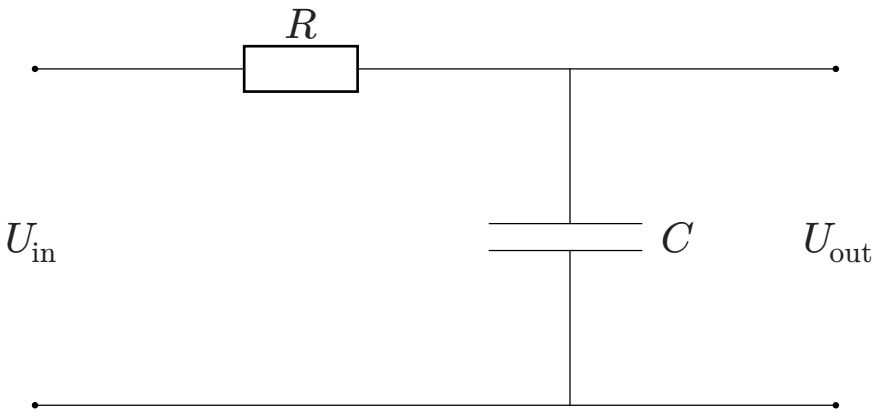


Figure 1: RC circuit

2.1 Low pass filter

We used

$$C = 100\ \text{nF} \qquad R = 1\ \text{k}\Omega \qquad U_{\text{in}} \approx 1.44\ \text{V} \tag{2.1}$$

And got the following results:

Ferquency f [Hz]	10	100	500	1000	5000	10000	50000	100000
U_{out} [V]	1.44	1.44	1.36	1.2	0.44	0.248	0.08	0.06
Phase shift φ [°]	0	−3.7	−17	−33.5	−73	−83	−89	−90

Table 1: Measured voltage U_{out} and phase shift between U_{in} and U_{out} for different frequencies

With the measured values from Table 1 we can make a graphical representation of $U_{\text{out}}/U_{\text{in}}$ of $\log(f)$.

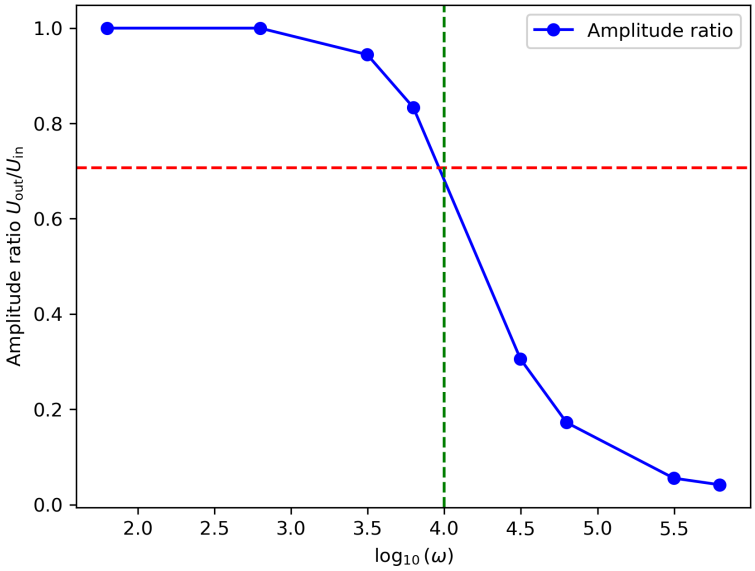


Figure 2: The x-axis shows the angular frequency $\omega = 2\pi f$ on a logarithmic scale. The y-axis shows the corresponding ratio $U_{\text{out}}/U_{\text{in}}$. The dashed red line is at $1/\sqrt{2}$. The dashed green line is approximately the intersection between the graph and the dashed red line.

In Figure 2 we can clearly see that up to a certain angular frequency the ration stays at approximately 1 and the drops to 0.

2.2 Cut-off-frequency

The cut-off-frequency is the frequency where the ratio of $U_{\text{out}}/U_{\text{in}}$ is $0 = \frac{1}{\sqrt{2}} \approx 0.707$ [1, p.11].

From Figure 2 we can estimate the cut of frequency at around:

$$\log(\omega) \approx 4 \cdot \log(\text{Hz}) \Rightarrow \omega \approx 10^4 \text{ Hz} \tag{2.2}$$

The theoretical value is [1, p.9]

$$\frac{U_{\text{out}}}{U_{\text{in}}} = \frac{1}{\sqrt{\omega^2 C^2 R^2 + 1}} \tag{2.3}$$

$$\Rightarrow |\omega_{\text{cutoff}}| = \frac{1}{|RC|} = 10^4 \text{ Hz} \tag{2.4}$$

This is exactly what we measured.

2.3 Limit cases

If $\omega \ll \omega_{\text{cutoff}}$ the ratio goes to 1 and for $\omega \gg \omega_{\text{cutoff}}$ the ratio goes to 0. This implies that only the low frequencies go through the circuit. This is exactly what it means to be a low pass filter.

2.4 Roll-off

The slope in Figure 2 equivalent to the "sharpness" of the filter is called the roll off. The roll off is given either in (dB / dec) or decibels per octave. Reading from the Figure 2 we get approximately

$$\begin{aligned}
10 \to 100 \text{ Hz} &\quad \Rightarrow \quad \text{roll-off} \approx 0.00 \text{ dB / dec} \\
100 \to 500 \text{ Hz} &\quad \Rightarrow \quad \text{roll-off} \approx -0.71 \text{ dB / dec} \\
500 \to 10^3 \text{ Hz} &\quad \Rightarrow \quad \text{roll-off} \approx -3.61 \text{ dB / dec} \\
10^3 \to 5 \cdot 10^3 \text{ Hz} &\quad \Rightarrow \quad \text{roll-off} \approx -12.47 \text{ dB / dec} \\
5 \cdot 10^3 \to 10^4 \text{ Hz} &\quad \Rightarrow \quad \text{roll-off} \approx -16.54 \text{ dB / dec} \\
10^4 \to 5 \cdot 10^4 \text{ Hz} &\quad \Rightarrow \quad \text{roll-off} \approx -14.06 \text{ dB / dec} \\
5 \cdot 10^4 \to 10^5 \text{ Hz} &\quad \Rightarrow \quad \text{roll-off} \approx -8.30 \text{ dB / dec}
\end{aligned} \tag{2.5}$$

2.5 Phase shift between U_{in} and U_{out}

With the numbers in Table 1 we can plot the phase shift between U_{in} and U_{out}.

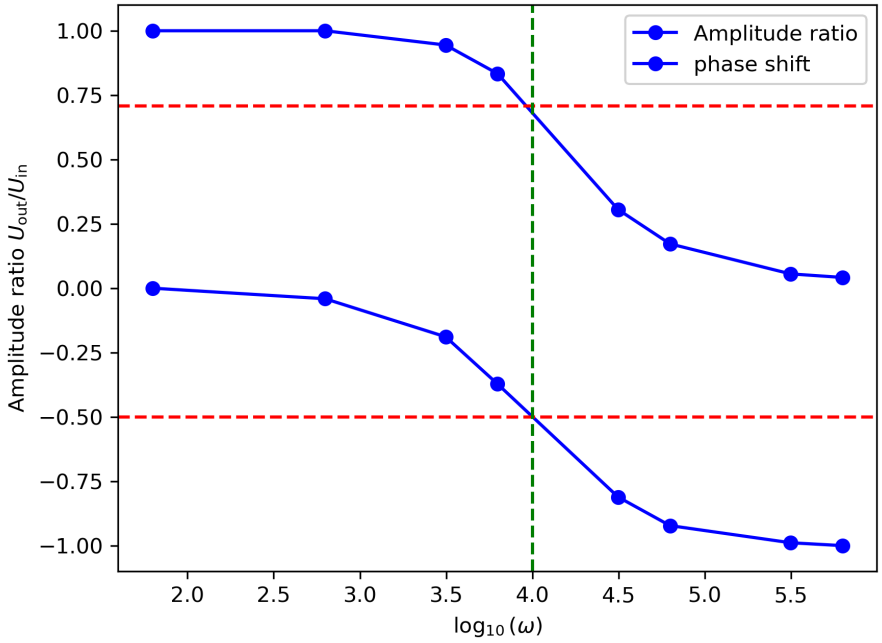


Figure 3: Added the normed phase shift between U_{in} and U_{out} in Figure 2

From there we can read off the $\varphi(\omega_{\text{cutoff}})$:

$$\varphi(\omega_{\text{cutoff}}) \approx -45° = -\frac{\pi}{4} \tag{2.6}$$

Some other phase shifts are in Table 1.

3 Part II

$L = 10mH, \quad C = 100nF, \quad R = 1k\Omega, \quad U_{\text{in}} = 1V \quad (2.8V_{\text{pp}})$

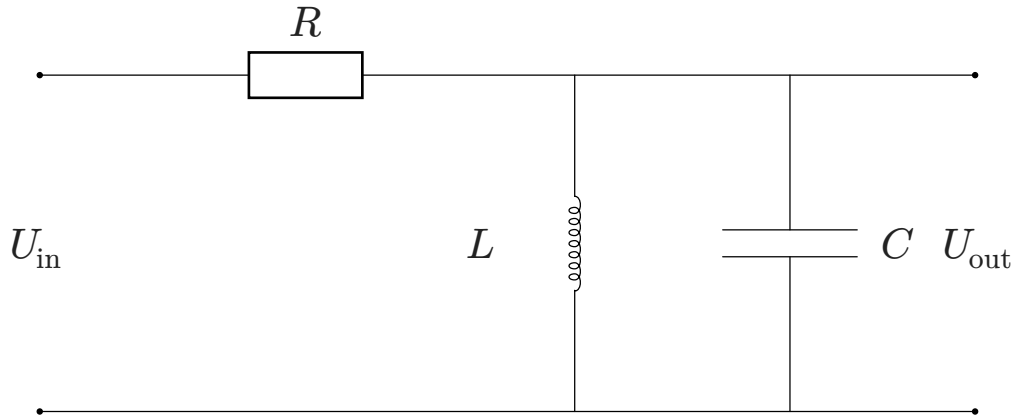


Figure 4: RLC circuit

3.1

for the phase shift between U_in and U_out we used math function

we also calculated:

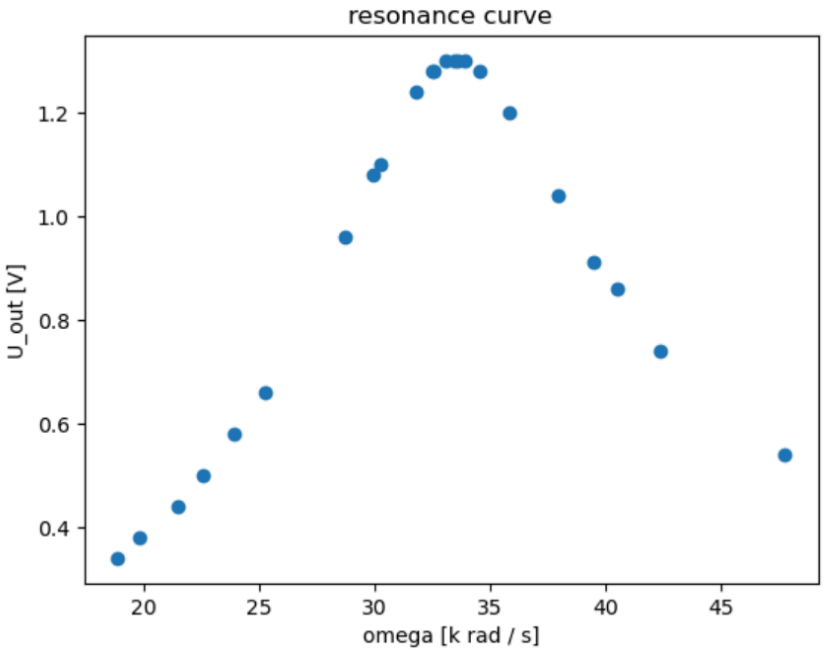
$$\frac{U_{\text{out}}}{U_{\text{in}}} = \left(R^2\left(\omega C - \frac{1}{\omega L}\right)^2 + 1\right)^{-\frac{1}{2}} \Rightarrow \omega_0 = \frac{1}{\sqrt{CL}} \approx 31'623\frac{\text{rad}}{s} \tag{3.1}$$

the measurements were:

Ferquency f [kHz]	3	3.15	3.42	3.59	3.81	4.02	4.57	4.76	4.81	5.18	5.4	5.5	5.7	6.04	6.28	6.45	6.74	7.6	5.32	5.26	5.35	5.06	5.17
U_{out} [V]	0.35	0.38	0.44	0.5	0.58	0.66	0.96	1.08	1.1	1.28	1.3	1.28	1.2	1.04	0.91	0.86	0.74	0.54	1.3	1.3	1.3	1.24	1.28
Phase shift φ [°]	72.6	71	67	65	62	58	40	32.4	30	8	−5.5	−11.5	−22.6	−38	−45	−49.2	−55	−65	−1.3	2.3	−2.5	16.6	9.5

Table 2: Measured voltage U_{out} and phase shift between U_{in} and U_{out} for different frequencies

3.2



$$\Rightarrow \omega_0 \approx 34000 \frac{\text{rad}}{s} \tag{3.2}$$

3.3

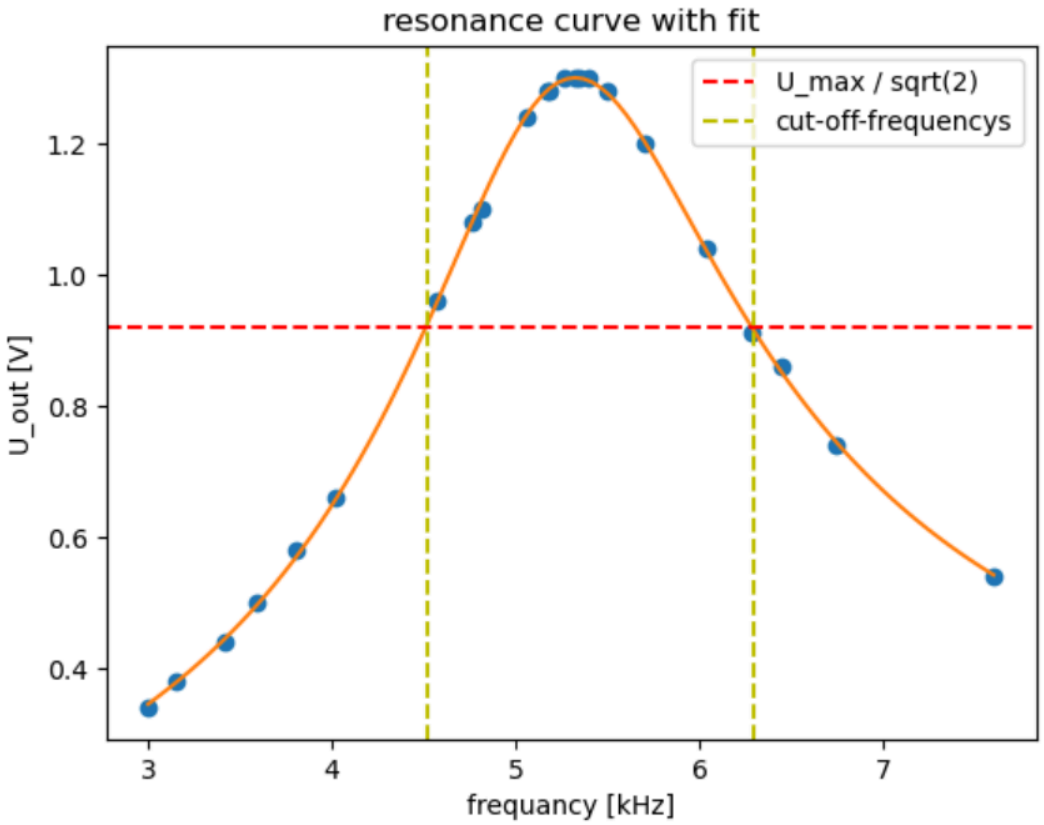


Figure 6: graphical Q: $Q_{\text{graphical}} = 2.99$

$$Q_{\text{theory}} = R\sqrt{\frac{C}{L}} \approx 3.16 \qquad [1] \tag{3.3}$$

since Equation (3.1) we used the following fit:

$$f(\omega, a, b, c) = \left(c + \left(\omega b - \frac{1}{\omega a}\right)^2\right)^{-\frac{1}{2}} \Rightarrow \omega_0 = \frac{1}{\sqrt{ab}} \tag{3.4}$$

and got the fit parameters: $a \approx 0.081, b \approx 0.43, c \approx 0.59$ where the inputvariable ω was used for kHz. the equation:

$$\frac{f(\omega_0, a, b, c)}{\sqrt{2}} = \frac{U_{\max}}{\sqrt{2}} \stackrel{!}{=} f(\omega, a, b, c) \tag{3.5}$$

gives the quadratic formular:

$$\omega^2 b - \omega\sqrt{\frac{2}{U_{\max}^2} - c} - \frac{1}{a} = 0 \tag{3.6}$$

this yields:

$$\omega_{\text{lower}} \approx 4.51 * 2 * \pi \quad k\frac{\text{rad}}{s}, \qquad \omega_{\text{upper}} \approx 6.23 * 2 * \pi \quad k\frac{\text{rad}}{s} \tag{3.7}$$

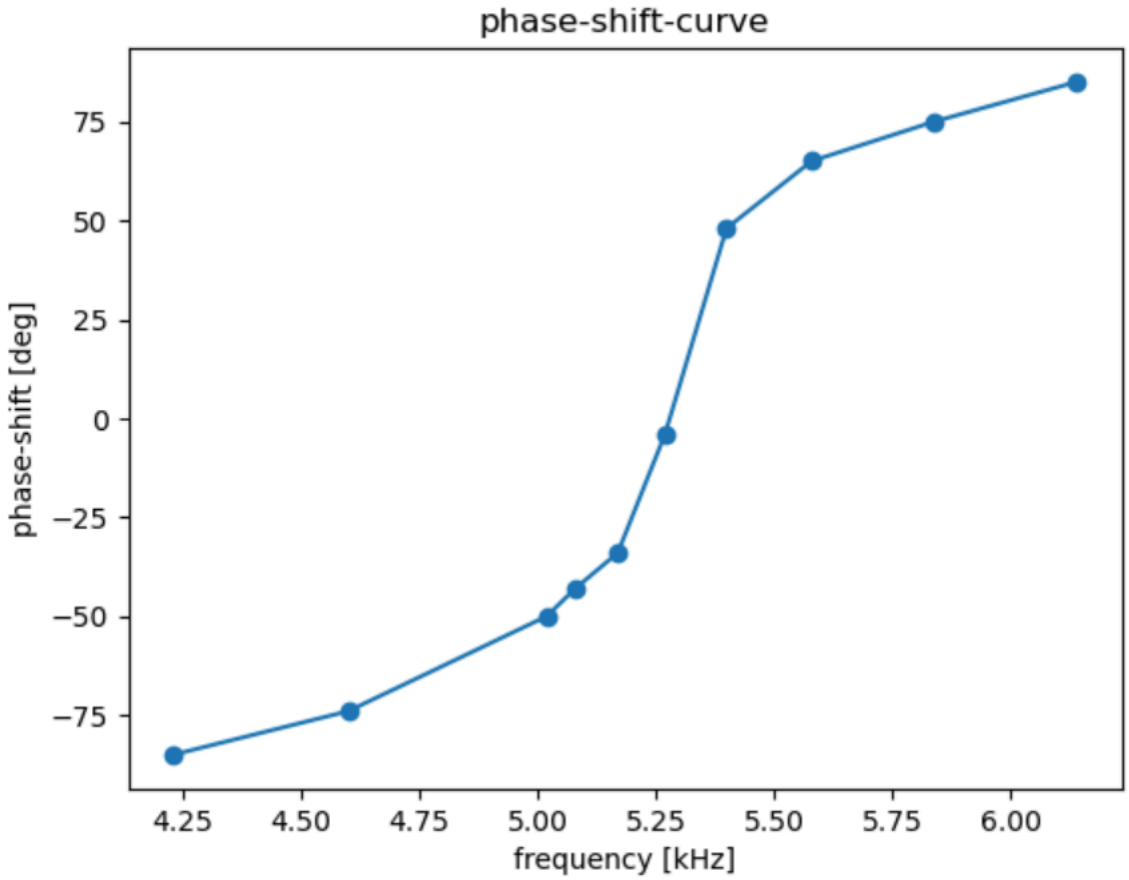
$$Q_{\text{fit}} = \frac{\omega_0}{\Delta\omega} = \frac{1}{\sqrt{ab}} \frac{1}{(\omega_{\text{upper}} - \omega_{\text{lower}})} \approx 3.00 \tag{3.8}$$

3.4

To measure the phase of the current I and U_{out} we measured the voltage over R (U_R), since it is in phase with the current. To measure U_R we did $U_{\text{in}} - U_{\text{out}}$ (ch-1 - ch-2). We did this with the math function on the oscilloscope. To get a reading for the phase we did phase beween math and ch-2, which is U_{out}.

Ferquency f [kHz]	4.23	4.6	5.02	5.08	5.17	5.27	5.4	5.58	5.84	6.14
I [A]	3.01	3.79	4.76	4.85	4.97	5.02	5	4.84	4.43	3.82
Phase shift φ [°]	−85	−74	−50	−43	−34	−4	48	65	75	85

Table 3: Current through the RLC-circuit and phase shift φ between current and voltage. U_{out}

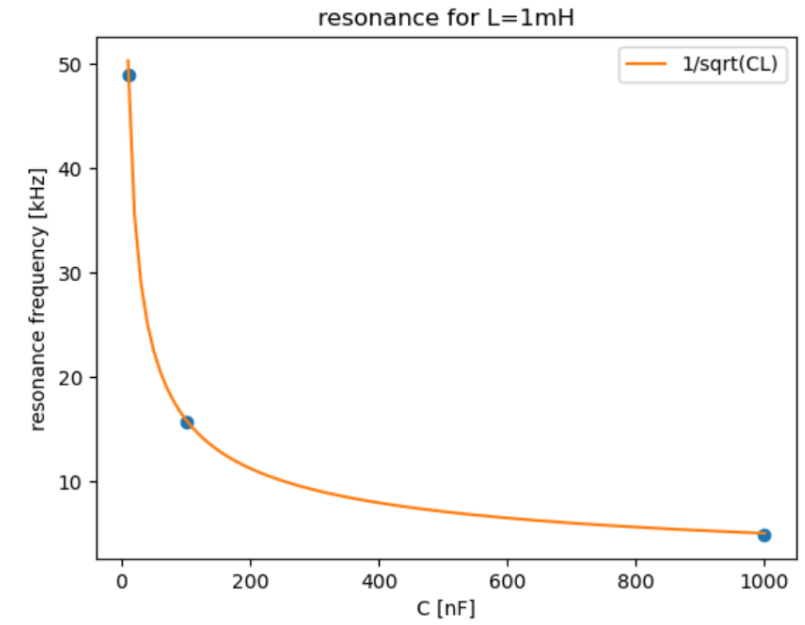


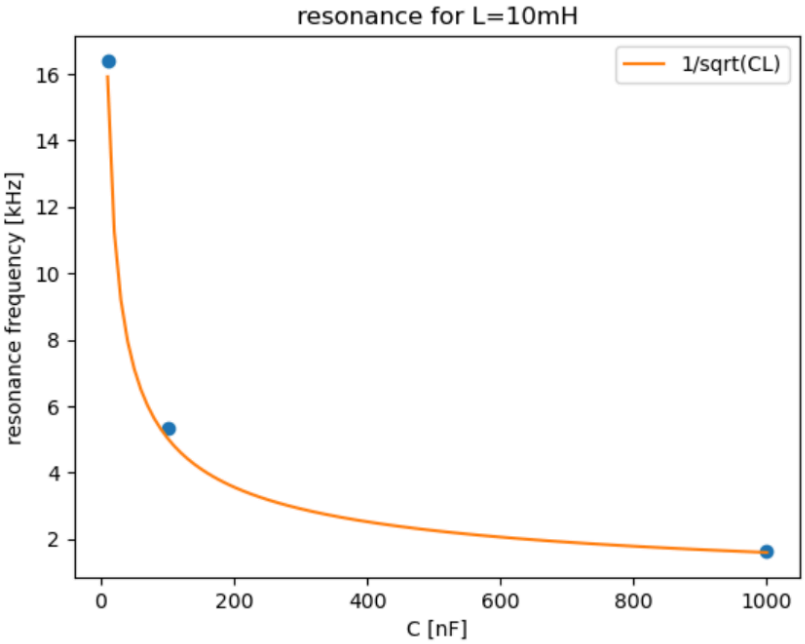
3.5

For the resonance frequencies at for different inductivities L and different capacities C, we measured:

(i)	$C = 10$ nF	$L = 1$ mH	$\omega_0 = 307k\frac{\text{rad}}{s}$
(ii)	$C = 100$ nF	$L = 1$ mH	$\omega_0 = 99k\frac{\text{rad}}{s}$
(iii)	$C = 1000$ nF	$L = 1$ mH	$\omega_0 = 31k\frac{\text{rad}}{s}$
(iv)	$C = 10$ nF	$L = 10$ mH	$\omega_0 = 103k\frac{\text{rad}}{s}$
(v)	$C = 100$ nF	$L = 10$ mH	$\omega_0 = 33k\frac{\text{rad}}{s}$
(vi)	$C = 1000$ nF	$L = 10$ mH	$\omega_0 = 10k\frac{\text{rad}}{s}$

Table 4: measured resonance frequencies at different L and C





from Equation (3.1) we get $\omega_0 = \frac{1}{\sqrt{LC}}$

3.6

At the resonance frequency (ω_0) we see sinus-shaped waves for both input signals. At 10 times the resonance frequency we see sinus-shaped waves, but with smaller Amplitude. The Fourier-Transform has peaks at odd multiples of resonance ω_0 ($\Rightarrow \omega_0, 3\omega_0, 5\omega_0, \ldots$) in addition the amplitude decreases each time. Because of the band-pass filter we can only see one of the peaks. The result is a sinus shape.

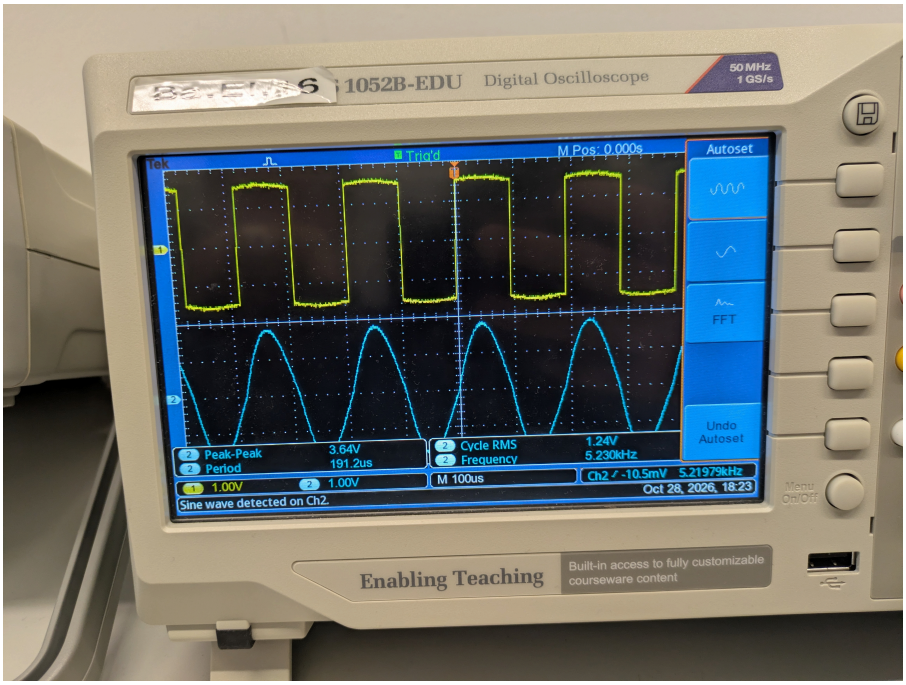


Figure 7: square wave at ≈ 5 kHz

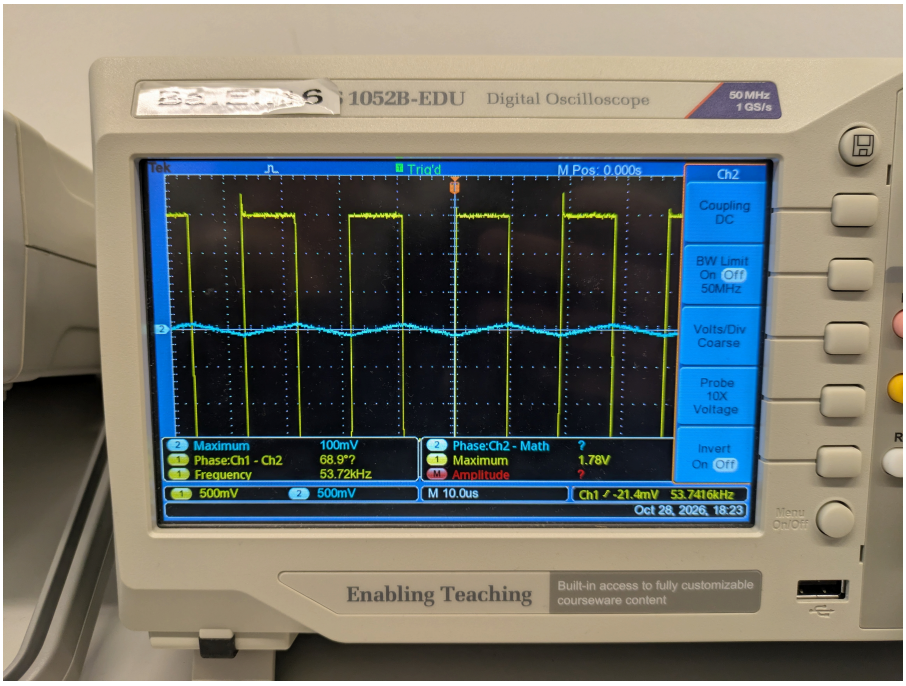


Figure 8: square wave at ≈ 50 kHz

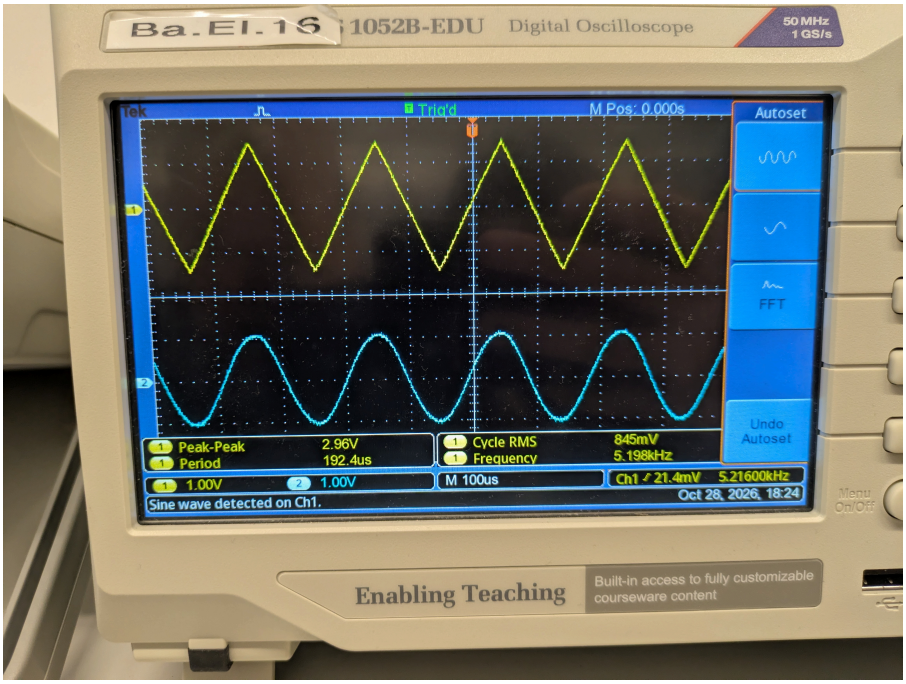


Figure 9: triangle wave at ≈ 5 kHz

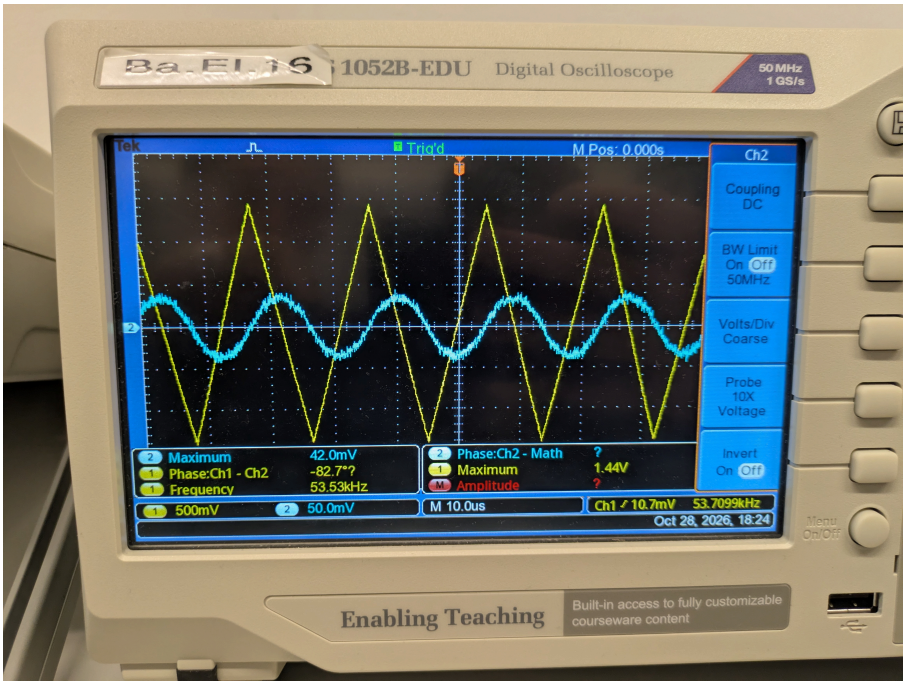


Figure 10: triangle wave at ≈ 50 kHz

Bibliography

[1] A. Eggenberger, T. Smolenski, and M. Kroner, "A simple state-of-the-art spectrometer for student labs: Cost-efficient, instructive, and widely applicable," *American Journal of Physics*, vol. 92, no. 2, 2024, doi: 10.1119/5.0164044.